Dongwook **Kwon** | AI Engineer ▪ Agentic AI Systems

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Experience

Suresoft Technologies **Seongnam-si, Korea**
AI Engineer, Intelligent Agent Technology Team *Dec. 2025 – Present*

- Developing an AI validation system utilizing AI agent–based architectures for enterprise model verification.
- Designing autonomous and collaborative AI agents that verify and evaluate machine-learning models at scale.
- Building agentic AI systems to enable systematic, scalable, and reliable AI verification processes.

Bio-Computing and Machine Learning Laboratory, Kwangwoon University **Seoul, Korea**
Research Assistant *Aug. 2021 – Dec. 2025*

- Researched deep-learning algorithms for anomaly detection in time-series data (ECG, cyber-physical systems).
- Co-developed an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.
- Published in EMNLP 2025 (Oral) and ACL 2026; co-author on two granted Republic of Korea patents.

LG Electronics Canada (with University of Toronto) **Toronto, ON, Canada**
Visiting Researcher *Jan. 2025 – Jul. 2025*

- Built an LLM-driven assessment framework for evaluating performance of agentic AI systems.
- Work fed directly into peer-reviewed publications at EMNLP 2025 and ACL 2026.

Qualcomm Institute, University of California San Diego **San Diego, CA, USA**
AI Development Project Participant, Summer Program *Jul. 2022 – Aug. 2022*

- Built a personality-based drug-addiction prediction system using machine-learning models.
- Recognized with the program's Achievement Award.

Education

Kwangwoon University **Seoul, Korea**
M.S. in Computer Engineering, GPA: 4.33/4.5 (98.1%) *Mar. 2024 – Feb. 2026*

University of Toronto **Toronto, ON, Canada**
Graduate Research Program, GPA: 3.68/4.0 *Jan. 2025 – Jul. 2025*

Kwangwoon University **Seoul, Korea**
B.S. in Electronics and Communications Engineering, GPA: 3.41/4.5 (88.1%) *Mar. 2019 – Feb. 2024*

Honors & Awards

Major Awards...

Aug. 2022: **Achievement Award**, Qualcomm Institute AI Development Project — UC San Diego *United States*

Dec. 2021: **Gold Award (Ministerial Award)**, 2021 Smart Maritime Logistics Project — Minister of Oceans and Fisheries *South Korea*

Academic Recognition

Nov. 2025: **Outstanding Student Paper Award**, Conference of the Institute of Electronics and Information Engineers *South Korea*

Nov. 2024: **Best Poster Award (Silver)**, 10th International Biomedical Engineering Conference (IBEC 2024) *South Korea*

Nov. 2023: **Outstanding Paper Award**, Conference of the Korean Society of Medical and Biological Engineering *South Korea*

Competitions

Dec. 2025: **Silver Award**, 23rd Embedded Software Competition (KIISE) *South Korea*

Dec. 2022: **Bronze Award**, Hanium ICT Mentoring Competition — FKII *South Korea*

Sep. 2022: **Excellence Award**, 18th Kwangwoon ICT Exhibition (KWIX) — Kwangwoon University *South Korea*

Oct. 2021: **Grand Prize**, MY (Multi-Y) Capstone Design Competition — Kwangwoon University *South Korea*

Funded Research Projects

LG Electronics Canada **Toronto, ON, Canada**
Agentic AI Evaluation Framework *Jan. 2025 – Present*
Developed an evaluation framework for agentic AI systems using LLMs to assess performance.

Sejong University **Seoul, Korea**
AI Intrusion Detection for APR1400 *Jul. 2022 – Present*
Developed an AI defense system to protect nuclear power plants from cyber attacks.

Ministry of Science and ICT **Seoul, Korea**
Quadruped Robot with Vision SLAM *Apr. 2022 – Nov. 2022*
Developed an autonomous quadruped robot using a navigation algorithm with SLAM and a depth camera.

Qualcomm Institute, University of California San Diego **San Diego, CA, USA**
Personality-Based Drug Addiction Prediction System *Jul. 2022 – Aug. 2022*
Developed a personality-based drug addiction prediction system using a machine learning model.

Kookmin University **Seoul, Korea**
Active Kill-Switch and Biomarker-Based Defense for IoT Medical Devices *Sep. 2021 – Dec. 2022*
Developed a security system for heart implant devices using an anomaly detection algorithm.

Publications

Journal Articles

2025: Y. Nam, J. Lee, H. Lee, **D. Kwon**, M. Yeo, S. Kim, R. Sohn, and C. Park, "Designing a remote photoplethysmography-based heart rate estimation algorithm during a treadmill exercise," *Electronics*, vol. 14, no. 5, p. 890, 2025.

2025: **D. Kwon**, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park, "Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments," under review at *Data Mining and Knowledge Discovery (Springer Nature)*, 2025.

Conference Papers

2026: R. Chang[†], **D. Kwon**[†], J. Lee[†], and N. Verma, "CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades," in *Proc. 64th Annu. Meeting Assoc. Comput. Linguistics (ACL) — Industry Track (Poster)*, 2026. [†]Equal contribution.

2025: J. Lee[†], R. Chang[†], **D. Kwon**[†], H. Singh, and N. Verma, "GEMMAS: Graph-based Evaluation Metrics for Multi Agent Systems," in *Proceedings of the EMNLP 2025 Industry Track (Oral)*, 2025. [†]Equal contribution.

2024: **D. Kwon**, M. Kim, Y. Kang, J. Lee, and C. Park, "Hybrid neural network model for anomaly detection in implantable devices using graph attention networks and transformers," in *Proc. 10th Int. Biomed. Eng. Conf. (IBEC)*, 2024. **Best Poster Award — Silver.**

2023: **D. Kwon**, Y. Kang, and C. Park, "Enhancing medical device security with GNN-GRU anomaly detection model," in *Proc. 62nd KOSOMBE Autumn Conf.*, pp. 204–205, 2023. **Outstanding Paper Award.**

Patents

Apr 2026: **Issued Patent**, "Method for Implantable Medical Device to Detect Anomaly in Real Time," KR 10-2961984.

Aug 2025: **Issued Patent**, "GPT API-Based Network Anomaly Detection," KR 10-2848814.

Leadership Experience

President, Kwangwoon International Student Association — Lead intercultural programs and student engagement initiatives.

Director of Filming and Editing, Kwangwoon Broadcasting Center (KWBC) — Oversaw media production and content delivery.

Class Representative, Dept. of Electronics and Communications Engineering — Represented student interests in academic affairs.

Executive Member, Amateur Radio Club — Organized radio workshops and technical exhibitions.